

Is the Effect of Organizational Ambidexterity and Technological Innovation Capability on Firm Performance Mediated by Competitive Advantage? An Empirical Research on Turkish Manufacturing and Service Industries

SAGE Open
October-December 2023: 1–21
© The Author(s) 2023
DOI: 10.1177/21582440231206367
journals.sagepub.com/home/sgo


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Abstract

The strategic management literature reveals that organizational ambidexterity (OA) and technological innovation capabilities (TIC) have separate effects on competitive advantage (CA) and firm performance (FP). However, these two capabilities have not been investigated simultaneously and in comparison with the sector. The main aim of this study is to reveal the mediating role of CA in the effect of OA and technological innovation capabilities on FP. Using the research model based on quantitative data, the primary data were collected by applying an online survey to 770 people selected by the random sampling method from the textile/ready-made garment/leather sector representing the manufacturing sector and the information technology sector representing the service sector working under the Istanbul Chamber of Commerce (ICOC) in Istanbul. The data collected were analyzed using SPSS. The results showed that CA mediated positive and statistically significant effects of OA and technological innovation capabilities on FP, and this mediating effect differed by sectors. The current study provided insights into manufacturing and service sector firms. Based on these results, recommendations for researchers and firm managers are presented in the conclusion section.

Keywords

organizational ambidexterity, technological innovation capability, competitive advantage, firm performance, resource-based view

Introduction

Firms perform all their activities to achieve their set goals. Performance is measured to determine the extent to which the goals are achieved in firms that are managed based on performance (Coşkun, 2006). The measuring of FP reveals insights into whether firms' strategies work. Achieving primary financial goals and becoming sustainable by providing a CA is essential for firms. This represents their capabilities to adapt to changing environmental conditions and their success (Bulut et al., 2009).

CA is a critical concept that helps us understand the process and outcomes of competition among competing firms in a sector (Dilek et al., 2020). Another

fundamental question of firm research is understand why FP differences occur. The basis for firm success is to obtain a CA when a firm outperforms its competitors (Knudsen et al., 2021). To create a CA, firms that will implement competitive strategies must have the resources and capabilities to implement these strategies and be able to predict which strategy will become successful under which conditions (Alayoğlu, 2010). The strategic

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management literature established the close relationship between resources and capabilities and CA through a RBV (Das & Teng, 2000). This view combines the perspectives of internal analysis, which has been the focus of many researchers since the mid-1980s, and external analysis, which has been the focus of previous approaches (Papatya, 2007). Valuable, rare, difficult-to-imitate or non-substitutable resources provide firms with a strategic advantage and above-average returns (Donnellan & Rutledge, 2019). According to the RBV, the competencies of firms performing better than other firms can be identified (Nielsen, 2002). Firms trying to fill these perceived gaps in resources and capabilities from the inside are more likely to create a CA (Donnellan & Rutledge, 2019). Accordingly, the source of the CA obtained by firms can be explained by the fact that the firm possesses more valuable resources than its competitors (McGrath et al., 1995). These elements form the basis of firms' strategies (Eren & Özdemirci, 2018).

In this respect, OA and technological innovation are among the prominent capabilities in the current literature. From a strategic management perspective, OA is a strategic capability that firms can pursue to maintain long-term business success, despite managerial challenges (March, 1991). Thus, the literature has discussed ambidexterity as a component required for the long-term survival of organizations. There are significant challenges in implementing and executing OA. On the one hand, supporters of exploration capability and alignment may resist attempts to increase discipline, hierarchy, and bureaucracy. On the other hand, supporters of exploitation capability and alignment may be reluctant to face the tensions and risks associated with uncertainty, instability, and change (Alpkan & Gemici, 2016). It is challenging to combine or balance these two approaches effectively (Soosay & Hyland, 2008). Ultimately, organizations that maintain balance become ambidextrous (Günsel et al., 2018). The purpose of ambidexterity is to secure an organization short-term and long-term CA (Clauss et al., 2021). From a resource-based perspective, a behavioral context in which exploitation and exploration capabilities can develop simultaneously can be considered a valuable, rare and costly resource to imitate and, thus, a potential source of CA (Simsek, Heavey et al., 2009). These research results on the RBV of strategic management support the importance of organizations ambidexterity capability.

TIC, another critical element, is considered the cornerstone of CA, especially for long-term strategic success. Firms with TIC gain a CA and grow as a result of their success (Figueroa & Conceição, 2000). According to Peteraf (1993), one of the advocates of the RBV, technological innovation capabilities are the primary sources of CA since they are sources with low mobility (Shan &

Jolly, 2013). This capability is a fundamental component of competitiveness embedded in a firm's products, services, processes and organizational structures (Gunday et al., 2011). J. C. Guan et al. (2006) state that improving TIC increases a firm's competitiveness (Wang et al., 2008). The capability to quickly introduce new products and adopt new processes is an essential aspect of competition. Hence, to adapt to the requirements of a firm's strategy, the special conditions of the market, and the competitive environment, the TIC should be defined in a broad scope (J. Guan & Ma, 2003). When the literature on the subject is reviewed in general, it is emphasized that the technological innovation capabilities of firms are an essential element in implementing the strategy and creating and maintaining CA (Zastempowski et al., 2020).

Based on the literature readings, it is thought that the mediation role of CA will make a significant contribution to the positive effect of OA and TIC on FP. Although studies on OA and TIC have increased since the 1990s, most research has focused on the concepts themselves and their unidirectional outputs. There have been limited recommendations regarding the impact of these capabilities on businesses, their performance, and their role in developing them. In addition, no study has been found that investigates the relationship of these variables with a CA and FP in the context of inter-sectoral comparison with high technology and medium technology intensity. To our knowledge, no theoretical or empirical study examines the effects of OA and technological innovation capabilities on FP using CA as a mediator variable. This study aims to contribute to the literature by revealing, with empirical findings, the mediating role of CA in the effect of OA and technological innovation capabilities on FP based on this gap in the literature. The issues as mentioned earlier represent important research gaps in the current literature. In this respect, the scope of the present study is expected to make a unique contribution to the literature in addition to its empirical results. To ensure generalizability to similar contexts, the current study aimed to bring more clarity by doing this comparatively in sectors with different technology intensities in a developing country. In this respect, the scope of the present study is expected to make a unique contribution to the literature in addition to its empirical results. This present study aims to answer the following research questions:

RQ1. What is the relationship between OA, TIC, CA and FP empirically?

RQ2. Does CA play a role as a mediator in the effect of OA and TIC on FP?

RQ3. Does the mediating role of CA in the effect of OA and TIC on FP vary according to sectors with different technology intensities?

From the point of view of implementers, academicians and firms can rely more on the study's findings, considering the data collected from different sectors. It provides managers with ways to understand the context's characteristics, increase OA and technological innovation capabilities and FP, and maximize organizational outcomes.

Literature Review

Organizational Ambidexterity

Organization researchers have adopted the term ambidexterity—the ability of individuals to use both hands equally well, as a metaphor to describe organizations (Lubatkin et al., 2006). Robert Duncan used this concept, for the first time in management and organizational studies in 1976 (O'Reilly & Tushman, 2008). According to Duncan (1976), organizations manage trade-offs between conflicting demands by establishing “dual structures” in which specific business units or groups within business units focus on alignment while others focus on adaptation (Duncan, 1976, p. 184; Gibson & Birkinshaw, 2004, p. 209). Ambidextrous firms have the capability to explore new opportunities in addition to their capability to use their available competencies (Kortmann et al., 2014). These firms can simultaneously implement two activities that seem opposite or different from each other (Cingöz & Akdoğan, 2015). They can perform exploratory activities for sustainable growth while using the existing business practices to maximize their returns (Clauss et al., 2021).

The study by March (1991) identified the dimensions of OA as exploitation and exploration capabilities (Gastaldi et al., 2022). While the exploitation capabilities dimension is defined as doing business by using the existing capabilities of firms, the exploration capabilities dimension is defined as doing business by using the capabilities that cover researching new information and technologies. Whereas exploitation capabilities are characterized by qualities such as “improvement, selection, production, efficiency, implementation, and execution,” exploitation capabilities are characterized by qualities such as “search, diversity, risk-taking, experimentation, play, flexibility, innovation, and discovery” (March, 1991, p. 71). In the literature, firms that implement Miles and Snow's defender strategy (1978) and Porter's (1980) cost leadership strategy are regarded as firms that implement strategies based on exploitation capabilities. Firms that implement Miles and Snow's (1978) prospector strategy and Porter's (1980) differentiation strategy are accepted as firms implementing exploratory strategies (Findıklı & Pinar, 2014; Hotz, 2010).

Another dimension distinction is between alignment and adaptability that Gibson and Birkinshaw (2004) introduced to the literature (Gibson & Birkinshaw,

2004). Among research in the field of OA, studies by March (1991), Benner and Tushman (2003), He and Wong (2004), Lubatkin et al. (2006), Menguc and Auh (2008), Raisch and Birkinshaw (2008), O'Reilly and Tushman (2013), Günsel et al. (2018), Clauss et al. (2021), and Wenke et al. (2021) adopted the approach based on exploitation and exploration capabilities while studies by Gibson and Birkinshaw (2004), De Clercq et al. (2013), and Kortmann et al. (2014) adopted the approach based on the distinction between alignment and adaptability (Bakan & Sezer, 2017; Kumkale, 2022). Although studies investigate OA with different dimensions, exploitation and exploration capabilities are the most widely used (Cheng & Van de Ven, 1996). OA was also examined under the dimensions of exploitation and exploration capabilities within the scope of this study.

The first studies in the business management literature suggested that it was impossible to improve exploitation and exploration capabilities simultaneously and argued that one of them should be focused on (McGill et al., 1992). In the following years, despite differences in organizational conditions enabling the existence of exploitation and exploration capabilities, researchers suggest a balance between the two can be established (He & Wong, 2004; Lubatkin et al., 2006; Tushman & O'Reilly, 1996). As a result, the OA hypothesis assumes that firms achieving a balance of exploitation and exploration capabilities acquire performance advantages that outperform their unwieldy competitors by putting emphasis on over-exploitation or over-exploration capabilities (Hughes, 2018). The literature also hypothesizes that successful organizations are ambidextrous. It is essential that these firms have both exploitation and alignment capabilities in managing today's demands and have exploration capability and adaptability in responding to changes in the environment (Duncan, 1976; Gibson & Birkinshaw, 2004; Jansen et al., 2005; O'Reilly & Tushman, 2013; Simsek, Heavey et al., 2009; Tushman & O'Reilly, 1996).

Technological Innovation Capability

Technological innovation is associated with changes in the foundation of a firm, including its primary business activities. It is about generating and applying new technological knowledge of how to do business differently and better in terms of a firm's products and services or operational processes (Heij et al., 2020). According to the OECD, Oslo Manual 3 (2005), interconnected product and process innovations are called technological innovations (Yıldız, 2018). Previous studies have conceptualized TIC as a multidimensional structure under the titles of product innovation capability and process innovation capability (Zastempowski et al., 2020). In this regard, the concept of TIC is “a firm's capability to make

interdependent product and process innovations.” Each product innovation triggers innovation in the processes using that product (Utterback & Abernathy, 1975, p. 645).

Conceptually, the product innovation dimension is developing a new or different version of a product or service and putting it on the market (Yılmaz, 2015). According to the OECD, Oslo Manual 4 (2018), “a product innovation is a new or improved product or service introduced to the market and differing significantly from the firm’s previous products or services” (OECD Oslo Manual 4, 2018, p. 21). Product/service innovations include improvements in technical specifications, components or materials, user convenience, and other functional properties (Tüzünkan, 2016). According to the OECD, Oslo Manual 4 (2018), the process innovation dimension is “a new or improved business process for one or more business functions made available by the firm and differing significantly from previous business processes of a firm” (OECD Oslo Manual 4, 2018, p. 21). Regarding the existing definitions, product innovation capability is a firm’s capability to develop new or significantly improved products. Process innovation capability is defined as a firm’s capability to develop new or significantly changed efficient technological processes (Yu et al., 2017). Process innovation is connected to product innovation and describes significant changes in the processes by which the product is created. Process innovations considerably impact production economies (Yılmaz, 2015). These innovations increase productivity and efficiency (Keser & Ve Çelik, 2020).

Various studies in the literature have attempted to find the elements of TIC (Siallagan et al., 2019). According to Lall, technological innovation capabilities internalize new knowledge to produce new processes and products resulting from organizational learning (Lall, 1992). Using the asset approach, Christensen (1995) explained the elements of technological innovation capabilities as scientific research, product, application, process, and esthetic design assets. Using the process approach, Chiesa et al. (1996) stated that technological innovation capabilities included organizational processes and activities. Technological innovation capabilities are formed by concept generation capability, product development capability, process innovation capability, technology acquisition capability, leadership capability, and capability to deploy resources and use systems and tools effectively (Ince et al., 2016). Burgelman et al. (2004) suggested TIC as an organization comprehensive features facilitating and supporting technological innovation strategies. Technological innovation capabilities are a remarkable asset of a firm, including critical areas such as technology, production, process, knowledge, experience, and organization (J. Guan & Ma, 2003; Tseng et al.,

2012). Shan and Jolly (2013) suggested technological innovation as firms’ capability to manage and organize resources to produce new technologies by selecting, acquiring, internalizing, and integrating new technologies (Shan & Jolly, 2013). Although the first studies in the literature have advocated the importance of external sources of TIC, subsequent studies have focused on internal capabilities; firms’ technological innovation capabilities are related to their capabilities to allow for the innovation process. A new product or service may be introduced, and related process innovations may be created due to the senior management decisions based on the firm’s innovation strategy.

Competitive Advantage

CA can be defined as a firm’s potential to consistently generate high profits against its competitors in the same market (Grant & Jordan, 2014, p. 174). As stated by Porter (1985), this advantage is obtained as a result of the strategies developed by a firm to protect and strengthen its targeted position in the sector in which it operates (Eroğlu & Yalçın, 2013). A firm has a “CA” if it can create more economic value than the marginal competitor in its market. This definition of the concept of CA provided by Barney (1986, J. Barney 1991) is consistent with its use by Porter (1985) and with the value-based approach toward CA presented by Peteraf (2001). According to Peteraf and Barney, the meaning of CA undoubtedly depends on a clear definition of what it means to “create economic value” (Peteraf & Barney, 2003, p. 314). If the value created in this sense is not higher and superior to its competitors, a firm does not gain a CA even if it can create value (Ülgen & Mirze, 2018). Researchers have different views on how a “CA” is achieved. According to one view, CA is achieved through “positional advantage in the market,” based on creating superior customer value or obtaining lower relative costs. According to another view, it is achieved by “firms’ core competencies” based on the relative superiority in skills and resources (Poth, 2014). The first of these views is called the “industry-based view,” whereas the second is called the “resource-based view (RBV)” (Barca & Esen, 2012).

According to the industry-based view developed by Porter, the external environment of firms is effective in determining the source of CA (Sönmez & Kasimoğlu, 2014, p. 70). To implement the industry-based view, Porter developed three related concepts: “five forces,” “generic strategies,” and “value chain” frameworks (Stonehouse & Snowdon, 2007, p. 257). To determine the CA in a specific industry, it is first necessary to define the firm’s value chain (Porter, 2001). A firm’s potential to be advantageous is negatively correlated with increased competition among firms, lower barriers to entry, a high

number of substitutes, and increased bargaining power of customers and suppliers. Based on the analysis of these five forces, he argues that a firm can develop a competitive strategy through an appropriate configuration and coordination of value chain activities (Stonehouse & Snowden, 2007). CA can develop in any area in the value chain (Ankli, 1992). As Porter (1980) stated, the two basic ways a firm can create a CA in any industry are cost leadership and differentiation strategies (Sigalas & Pekka Economou, 2013, p. 64). Firms implementing the cost leadership strategy to gain a CA face higher demand levels and price sensitivity in their sector (Gómez et al., 2021). Therefore, they aim to become the lowest-cost manufacturer. Benefiting from economies of scale is the most widely adopted way for firms to implement this strategy (Peker et al., 2016). Firms can take advantage of the differentiation strategy if they can introduce to the market products and services that are considered unique across the sector (Porter, 2015, p. 45). They differ from their competitors when they offer a product or service of unmatched value to their buyers rather than offering it at a low price. A differentiation advantage occurs when firms obtain a high price that will exceed the cost of differentiation in the market through differentiation (Grant & Jordan, 2014). Firms with this advantage can both demand high prices for their products or services and gain a market share from their competitors by increasing the general demand for these products and services (Hill et al., 2019). Both the differentiation and cost leadership strategies can be implemented in all value-creating activities of the firm. Thus, firms can gain an advantageous position against the five competitive forces (Porter, 2015, p. 45).

Contrary to the industry-based view, the RBV investigates which features in the internal structures of firms provide them with a CA (Eren & Özdemirci, 2018). This view has an organizational perspective that evaluates the organization effectiveness by observing how successful firms acquire, bring together and manage valuable resources (Daft, 2015). In this internal analysis, emphasis is put on differences between resources and capabilities. Capabilities are obtained due to firms' resources to create knowledge and skills (Papatya, 2007). In addition to the assumption that firms try to maximize their profits and the limited rationality assumption, the RBV defends the assumptions of resource heterogeneity and resource immobility (J. B. Barney & Ve Arıkan, 2001). The assumption of resource heterogeneity assumes that firms within a sector have different resources and can be heterogeneous according to the resources they control (Fidanboy & Sargut, 2021). Since the skills and capabilities of firms differ, they can gain a CA using different sets of resources they have. In the same external environment and under market conditions with perfect

competition, firms can thus achieve relatively better performance than each other (Nothnagel, 2008). According to the assumption of resource immobility, heterogeneity may last long since the resources controlled by firms within a sector cannot be utterly mobile among firms. According to this assumption, since resources are immobile, they cannot be easily obtained by other firms, and the same strategies cannot be implemented. Hence, a firm's CA is sustainable to the extent that its resources are heterogeneous and immobile (J. Barney, 1991). These resources, which can provide a CA to firms, are valuable, rare, inimitable, and non-substitutable. Firms create their core competencies using these resources (Ülgen & Mirze, 2018). In conclusion, researchers have discussed that the content and process of CA strategies should be considered together, in addition to the two separate views explaining CA. While the industry-based advantage view adapts most quickly to the analysis of the strategy content, the RBV adapts more easily to the strategy process issues (Reed et al., 2000).

Firm Performance

FP can be defined as an indicator of the extent to which a firm can fulfill its establishment purposes (Bayyurt, 2007). Along with the concept of FP, the understanding of the FP dimensions has changed and followed a developing process until the present day. Within this process, considering performance as a multidimensional concept has become widespread. McGivern and Tvorik (1997) argued that the dimensions of FP could be assessed on financial and non-financial dimensions under any conditions. A financial perspective stresses the importance of external market factors such as the firm's competitive position. A non-financial perspective is established on behavioral and sociological paradigms and their compatibility with the environment (McGivern & Tvorik, 1997). Likewise, Ngo and O'Cass (2013) also defined FP as assessing a firm's success in the sector through financial and non-financial indicators (Chaithanapat et al., 2022). Financial performance, expressed by fulfilling economic targets, is measured numerically by profitability, sales growth, and earnings per share (Ozmutlu & Can, 2022). The non-financial performance involves market share, new product presentation, product quality, marketing effectiveness, adding value to production, and technological activities (Zehir, 2016). The financial performance dimension is the dimension measuring the profitability of strategies (Koçel, 2020). This change indicates the firm's level of achievement of its economic goals according to the targeted profitability and growth (Kaplan & Norton, 1992). This dimension is widely used throughout the development of the concept of FP because it provides precise and objective information (Coşkun, 2006). The

dimension, addressed under the name of growth performance, shows market share and sales revenues (Koçel, 2020). Factors such as position in the competition and general profitability level, in addition to factors such as growth in the number of products offered to the market, growth in market share, growth in the number of employees, and growth in the number of new customers, indicate growth in FP (Ağca, 2009; Baker & Sinkula, 1999; Venkatraman, 1989). When measuring growth performance, it is vital to consider not only a firm's change but also its position relative to its competitors. Some of the factors that the financial performance dimension cannot measure are measured with this dimension (Coşkun, 2006). Many studies on performance measurement have demonstrated that FP is assessed with the financial performance and growth performance dimensions (Altındag et al., 2011; Antoncic & Hisrich, 2001; Venkatraman, 1989). Within the scope of the present study, the dimensions of FP were addressed quantitatively with the financial performance and growth performance dimensions based on the study by Altındag et al. (2011).

Creating Research Hypotheses

This section includes a discussion of the theoretical background on which the research design of this study is based and the creation of the research hypotheses in this context. The RBV provides a comprehensive basis for understanding competition in modern markets and has inferences about how to cope with competitive challenges based on organizational and technological capabilities (Nielsen, 2002). In this regard, the RBV, focusing on the importance of the resources and capabilities of firms, formed the basis for addressing the variables and main hypotheses in creating the study's theoretical background. Additionally, the industry-based view was used to measure CA, considering its facilitating role.

The Relationship Between Organizational Ambidexterity (OA) and Firm Performance (FP)

Researchers have expressed contradictory views on OA and its impacts on FP in the literature (Kassotaki, 2022). Gibson and Birkinshaw (2004) found that obtaining ambidexterity through contextual support is possible, which positively correlated with FP (Gibson & Birkinshaw, 2004). He and Wong (2004) argued that OA had a direct positive effect on FP, but it was not substantial. Since this study assessed FP as the sales growth rate, the results indicated versatile firms, that is, firms implementing exploration and exploitation strategies together, achieved higher FP (He & Wong, 2004). Raisch and Birkinshaw (2008) conceptually presented the antecedents, moderators, and consequences of OA. In the

model they created by summarizing the most important studies, they determined that the most studied FP dimensions in the studies investigating the relationship between OA and FP were financial performance, market performance, and growth performance (Raisch & Birkinshaw, 2008). As a result of their empirical research, Han and Celly (2008) showed that international startups with ambidexterity capability outperformed those without such capability (Han & Celly, 2008). In his meta-analysis study, Simsek (2009) argued that environmental dynamism and complexity moderated the relationship between OA and FP. Also, the effect on FP would be more substantial in the case of the high moderation relationship (Simsek, 2009). Junni et al. conducted a meta-analysis of previous studies on OA and FP. Their findings revealed that OA was especially important for performance in non-manufacturing industries and at higher levels of analysis (Junni et al., 2013). In their study conducted in Turkey, Altındağ and Bilaloğlu Aktürk (2020) concluded that ambidexterity does not affect company performance (Altındağ & Bilaloğlu Aktürk, 2020). The following hypothesis was developed based on these previous studies:

H₁: Organizational ambidexterity (OA) positively impacts firm performance (FP).

The Relationship Between Organizational Ambidexterity (OA) and Competitive Advantage (CA)

Hamel and Prahalad (1993) found that achieving OA, which had strategic importance in gaining a CA for firms, was possible by catching the balance between exploration and exploitation strategies (Tunç, 2017). Tushman and O'Reilly's (1996) conceptual study revealed that managers and firms must be versatile and able to implement exploitation and revolutionary exploration strategies together to be successful over long periods (Tushman & O'Reilly, 1996). In their study carried out in 2008, Han and Celly showed that international new business ventures building their dynamic capabilities by developing versatile strategic skills could create long-term sustainability by gaining CA and superior performance (Han & Celly, 2008). Amniattalab and Ansari (2016) supported their research by finding that among OA and CA, the strategic foresight capability enabled firms to create unique resource combinations before their competitors in versatile innovation practices (Clauss et al., 2021). As a result of the study conducted by Bakan and Sezer (2016) and in which scale reliability was relatively high, the researchers found a significant and positive relationship between OA and competitive strategies (Bakan & Sezer, 2016). In their study on the German engineering industry in 2020, Clauss et al. found that

exploration strategies were effective in CA. However, exploitation strategies alone were ineffective unless they were combined with agility (Clauss et al., 2021). The hypothesis below was developed based on the said studies:

H₂: Organizational ambidexterity (OA) positively impacts competitive advantage (CA).

The Relationship Between Technological Innovation Capability (TIC) and Firm Performance (FP)

There are many studies suggesting that there is a positive relationship between innovation and FP (García Manjón et al., 2016). He and Wong (2004) found a positive relationship between TIC and FP (He & Wong, 2004). In the study carried out by J. Guan and Ma (2003), interdependence between the total development of innovation capability and export growth was found (J. Guan & Ma, 2003). Yam et al. (2011) investigated the effect of TIC between regional innovation performance and the innovation performance of firms in the region. They confirmed the mediator role played by technological innovation capabilities with empirical evidence. They determined that resource allocation, production and organizational capabilities were positively related to a firm's sales performance among all technological innovation capabilities (Yam et al., 2011). In their study (2012), Camison and Lopez analyzed the relationship between organizational innovation and technological innovation capabilities on FP. The study confirmed that organizational innovation supported the development of technological innovation capabilities, and both organizational innovation and technological capabilities for products and processes led to superior FP (Camisón & Villar-López, 2014). The study by Yang (2012) showed that TIC was related to long-term firm growth. The findings emphasized the importance of innovation intention and infrastructure for a firm's innovation capability (Yang, 2012). In the study investigating the effects of TIC on FP, Azubuike (2014) confirmed the presence of a relationship between FP in the development of new products with technological innovation (Azubuike, 2014). In their empirical study on manufacturing firms, Chen et al. (2020) revealed the mediating role of TIC between organizational innovation and FP (Chen et al., 2020). Ferreira et al. (2020) recently found a positive relationship between innovation capability, FP and CA (Ferreira et al., 2020). YuSheng and Ibrahim (2020) also found a positive and significant relationship between product and process innovation capabilities and FP in their study in the developing country of Ghana (YuSheng & Ibrahim, 2020). The following hypothesis was developed based on the said studies:

H₃: Technological innovation capability (TIC) positively impacts firm performance (FP).

The Relationship Between Technological Innovation Capability (TIC) and Competitive Advantage (CA)

Freeman (1994) stated that the primary source of CA was TIC in the long run (Freeman, 1994). Yam et al. (2004) confirmed that R&D and resource allocation capabilities were the two most significant technological innovation capabilities that affect FP. Whereas the resource allocation capability increases sales growth in small firms, a strong R&D capability makes it possible to maintain the innovation rate and competitiveness of the product in large and medium-sized firms (Yam et al., 2004). The study by J. C. Guan et al. (2006) on a sample of innovative firms from different industries showed that there were many areas for firms to improve their competitiveness when TIC and competitiveness scores were limited in the relationship between TIC and firm competitiveness (J. C. Guan et al., 2006). Betz (2011) determined that technological innovation affected the future competitiveness of a firm. A newly developed technology provides a CA for a technology leader firm and creates the need for innovation as a defensive position for a technology-follower firm (Betz, 2003). In their study on manufacturing firms, Liu and Jiang (2016) found a relationship between TIC and product competitiveness. Their study revealed that the firm's strategy capabilities, information resources, application R&D and manufacturing capabilities significantly impact new product development performance and product competitiveness (Liu & Jiang, 2016). Strand et al. (2017) determined that production and marketing capabilities significantly affected FP among the seven dimensions of TIC put forward by Yam et al. (2004) on an industry cluster (Strand et al., 2017). The study by Saunila (2020) found that the innovation capability in the context of small firms provided the basis for conscious organizational actions to develop innovation output and for firms to achieve sustainable CA by reviewing the empirical literature (Saunila, 2020). The hypothesis below was developed based on these studies:

H₄: Technological innovation capability (TIC) positively impacts competitive advantage (CA).

The Relationship Between Competitive Advantage (CA) and Firm Performance (FP)

Pelham and Wilson (1995) found that increasing the cost leadership strategy significantly and adversely affected the change in new product success and significantly and positively affected the change in growth. No significant

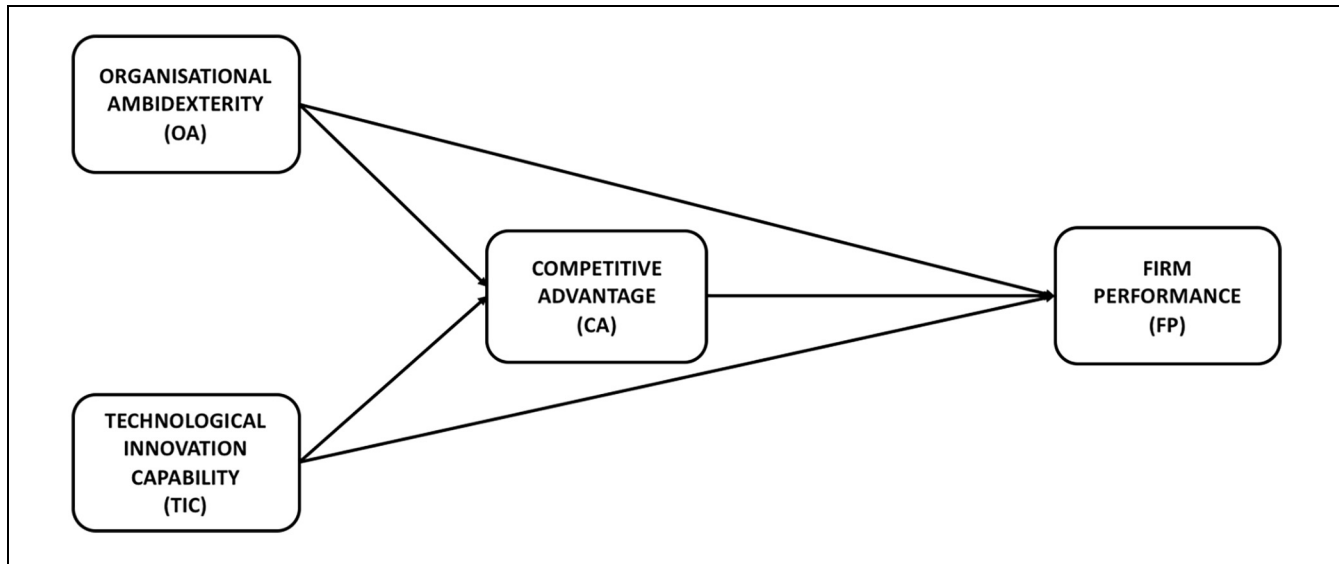


Figure 1. Research framework.

effect could be found on changes in any of the performance variables since the differentiation strategy needed to be implemented over a long period for changes in this strategy to affect performance. In their study, both strategies did not have a significantly impact profitability (Pelham & Wilson, 1995). Ma (2000) argued that CA was not equal to superior performance, and CA is context-specific (Ma, 2000). In their empirical research on export firms, Albaum and Tse (2001) found that a firm's performance was positively correlated with its CA in its marketing strategy, as the firm's CA determines its performance. Furthermore, they suggested that CA links strategy adaptation and FP (Albaum & Tse, 2001). Chan et al. (2004) failed to find an effect of high-performance human resources on FP. They suggested a low-level positive relationship exists between some dimensions of organizational culture and FP. The moderating role of the differentiation strategy in the relationship between high-performance human resources practices and FP was also not found (Chan et al., 2004). Akman et al. (2008) revealed significant and positive relationships between innovative, offensive and defensive strategies and FP as a result of their empirical research on manufacturing firms. The type of strategy that affects FP most is innovative strategy. Moreover, being technology-oriented also positively impacts FP (Akman et al., 2008). In their empirical study, Peker et al. (2016) determined that, among Porter's generic strategies, the cost leadership strategy affected customer-oriented performance, sales and profitability performance, while the differentiation strategy affected sales and profitability performance at an average level and strongly affected customer and product performance (Peker et al., 2016). The hypothesis below was developed based on the mentioned studies:

H₅: Competitive Advantage (CA) positively impacts Firm Performance (FP).

Mediating Relationships

Mediation hypotheses are based on the fact that there is another variable that interferes with the relationship between independent variables and dependent variables (Mwakapala & Sun, 2020). In light of the relationships and findings of all the research discussed above, the following hypotheses were developed regarding mediating relationships.

H₆: Competitive advantage (CA) plays a mediating role in the effect of organizational ambidexterity (OA) on firm performance (FP).

H₇: Competitive advantage (CA) plays a mediating role in the effect of technological innovation capability (TIC) on firm performance (FP).

H₈: The mediating role of competitive advantage (CA) in the effect of organizational ambidexterity (OA) on firm performance (FP) is different in the manufacturing and service sectors.

H₉: The mediating role of competitive advantage (CA) in the effect of technological innovation capability (TIC) on firm performance (FP) is different in the manufacturing and service sectors.

Research Design

A quantitative research method was used to test the accuracy of the hypotheses described above. In this approach, one of the aims is to explain the studied subject through the relationships between the independent, dependent,

and mediator variables (Creswell, 2017). In this respect, the independent variables of the current study, which was conducted to investigate the mediating role of CA in the effect of OA and TIC on FP, were determined as OA and TIC, and the dependent variable was determined as FP. Furthermore, CA, which was thought to strongly affect the effect of OA and TIC on FP, was considered a mediator variable.

The conceptual model of the research given in Figure 1 below was developed using the relationships in the literature.

Research Method

The analysis of the research data concerning the scales and the sample consists of two parts: the preliminary application and the main application. Validity, reliability, and factor analyses were performed in the preliminary application. In the main application, validity and reliability and factor analyses were performed, mean and standard deviation values were calculated, and correlation analysis was carried out. Subsequently, it was revealed whether the hypotheses were supported using regression analysis. SPSS was used within the framework required by the analysis.

Measures and Sampling

Measures. The scales with high validity that were used in international studies were selected. The questionnaire, created following the research design using the selected scales below, was applied to the participants in a 5-point Likert system. The details of the scale items used in the questionnaire are presented in the factor analysis table (Supplemental File, Table A1).

OA. For the OA scale, a two-dimensional scale from the study by Lubatkin et al. (2006) that consisted of 12 questions, six of which measure exploration capabilities and six measure exploitation capabilities, was used.

TIC. The innovation scale created by Vila and Kuster (2007), whose validity and reliability was tested and widely used in the literature, was chosen as the technological innovation scale. The section of this scale, consisting of four main dimensions and 24 statements regarding the product and process innovation dimensions, defined as technological innovation, was used within the scope of this study. The product innovation scale consisted of five statements, while the process innovation scale consisted of 11 statements.

CA. The scale from the study conducted by Zehir (2016) was used for the CA scale, and the scale adapted

from the studies by Dess and Davis (1984), Porter (1980), Slater and Narver (1993) was used for the cost leadership strategy. In addition, the two-dimensional scale, each of which consists of 14 questions, adapted from the studies by Kohli and Jaworski (1990), Lynch et al. (2000), Dess and Davis (1984), and Porter (1980), was used for the differentiation strategy variable.

FP. The 12-item scale from the study by Altındag et al. (2011) was employed to measure FP. This scale was adopted by the researchers based on the studies by Zahra et al. (2002), Lynch et al. (2000) and Baker and Sinkula (1999). On this scale, FP was addressed as two dimensions, financial performance and growth performance, and each sub-dimension consisted of a six item scale (Altındag et al., 2011).

Sample and Procedure. The study focused on two separate sectors, the manufacturing and service sectors. The study population consisted of firms in the textile/ready-made garment/leather and information technology sectors operating under the Istanbul Chamber of Commerce (ICOC) in the Istanbul province. Since it was impossible to reach all the registered firms, the study's sample size was calculated at a 95% confidence interval and determined in the following way. The population's sample size was calculated 380 since the number of firms operating under the ICOC in the Ready-Made Garment/Textile/Leather sector was 31,795. Because there were 15,114 operating under the ICOC in the Information Technology sector, the population's sample size was calculated as 375. In the current study, the total number of questionnaires collected and accepted as valid is 770. The sectoral distribution of the collected questionnaires is 386 for the Ready-Made Garment/Textile/Leather sector and 384 for the Information Technology sector. In terms of sample size, these numbers are acceptable as they are above the adequate sample size calculated at the beginning of the study.

To achieve the aims of the study, the survey method used extensively in social sciences research was employed in data collection. Data collection was performed by using an on-line questionnaire, created in Google forms. The online questionnaire consisted of 80 questions, of which 12 were demographic, and 68 were scale questions. Before distributing the questionnaire to all firms, a pilot study with 100 participants was conducted. The questionnaires were discussed with the firms' managers in advance before the online questionnaires were delivered. The validity and reliability analyses of the scales used were conducted on the data obtained from these 100 questionnaires. According to the results, Cronbach's alpha was calculated as .906 for the OA Scale, .926 for the TIC Scale, .947 for the CA Scale, and .911 for the FP

Scale. Moreover, the sub-dimension reliability was in the range of 0.777 to 0.939. All results were observed to be quite reliable and there was no need to cancel any questions from the questionnaire applied at the pilot stage. The questionnaire was then delivered to a larger sample population, and results were collected for 6 months. When the number of data samples reached 777, the survey was stopped, and the data analysis was initiated. Since outliers and extreme values increase the value of error variance and affect the power of statistical tests, outliers were examined before statistical tests, and it was checked whether they were present in the data sets. Outliers and extreme values were determined by the “Mahalanobis” method, and multiple normality criteria were provided by removing them from seven data sets.

As statistical remedies proposed by Podsakoff et al. (2012), we sought to prevent common method bias. The Harman’s single-factor test was used, in which all variables are loaded into one common factor (Podsakoff et al., 2003, 2012). A total variance of less than 50% for a single factor (in this study, 30.4%) indicated that common method bias does not affect the current study data (Harman, 1960).

Considering the distribution of the participant’s demographic characteristics, 52.2% were female, and 47.8% were male. Of the participants, 60.5% were married, and 39.5% were single. Regarding their educational status, 60.6% of the participants held a bachelor’s degree, 22.3% held a master’s degree, 7.1% were high school graduates, 7.1% held an associate degree, 2.1% held a PhD, and 0.6% were primary school graduates. Of the participants, 50.1% were mid-level managers, 37.4% were senior managers, and 12.5% were firm owners. The mean total working years of the individuals was 19.33 ± 12.36 , and the working time in the last institution was 8.76 ± 7.48 . Upon examining the boundaries of the field of activity of the firms where individuals worked, 74% were international. Concerning the sector distribution, 49.9% participated in the IT sector, and 50.1% participated in the textile/ready-made garment/leather sector.

Measure of Validity and Reliability

Reliability analysis was performed to test whether the statements in the scales were consistent and whether all the statements measured the same subject (Pallant, 2020, p. 113). For the results to be reliable, the measurements must be reliable. In this respect, the scale’s reliability was examined with Cronbach’s alpha. In the reliability analysis, if Cronbach’s alpha (α) coefficient value varying between 0 and 1 is .00 and .40, the scale is not reliable; between .40 and .60, the scale has low reliability; between .60 and .80, the scale is quite reliable; and between .80

Table 1. Reliability of the Scales and Dimensions Used in the Study.

Scales and dimensions	Item number	Cronbach’s alpha
Organizational ambidexterity (OA)	12	.863
Exploitation capabilities	6	.820
Exploration capabilities	6	.894
Technological innovation capability (TIC)	16	.908
Product innovation	5	.841
Process innovation	11	.814
Competitive advantage (CA)	28	.927
Cost leadership strategy	14	.948
Differentiation strategy	14	.946
Firm performance (FP)	11	.909
Financial performance	6	.939
Growth performance	5	.803

and 1.00, the scale is highly reliable (Tavşancıl, 2005, p. 19). For this study, as seen in Table 1, Cronbach’s alpha was calculated as .863 for the OA Scale, .908 for the TIC Scale, .927 for the CA Scale, and .909 for the FP Scale, which indicated all scales used were highly reliable. Furthermore, the sub-dimension reliability was found to be in the range of 0.803 to 0.948, thus, highly reliable.

Before applying the exploratory factor analysis, the Kaiser-Meyer-Olkin (KMO) test and Bartlett’s test of sphericity were applied to test the suitability of the sample size and item number for factorization. KMO values should be between 0 and 1 (Kaiser, 1974). In Bartlett’s test of sphericity, the obtained chi-square values should be significant ($p < .05$) (Hair et al., 2014). KMO values for our scales were found between 0.860 and 0.950, and when Bartlett’s test of sphericity was examined, the obtained chi-square values were significant ($p < .01$). It was concluded that the sample size and items were adequate for factor analysis. Accordingly, it was assumed that the data came from a multivariate normal distribution. After the suitability of the data for factor analysis was confirmed, exploratory factor analysis was carried out using Principal Components Analysis and Varimax rotation methods to examine the scale’s factor structure. The explained variance ratio showing the total variability was calculated. The results were as follows: For the OA, $KMO = 0.860$; ($\chi^2(66) = 4568.107$ $p < .01$; Total variance explained = 59.734%. For the TIC, $KMO = 0.870$; $\chi^2(120) = 6955.562$ $p < .01$; Total variance explained = 57.517%. For the Competitive Strategy, $KMO = 0.950$; $\chi^2(378) = 15953.049$ $p < .01$; Total variance explained = 60.283%. For the FP, $KMO = 0.923$; $\chi^2(66) = 5942.710$ $p < .01$; Total variance explained = 69.657%. The factor loadings of scale number 68 in the FP Scale were excluded from the analysis because they remained outside the theoretical dimensions (factor loading < 0.50). As a result of the analyses, all scales

Table 2. Descriptive Analysis, Kurtosis-Skewness Values and Pearson's Correlation Analysis.

Variable	N	Mean	SD	Kurtosis	Skewness	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Organizational ambidexterity (OA) (1)	770	4.38	0.47	1.970	-1.393	1											
Exploitation capabilities (2)	770	4.34	0.5	0.155	-0.597	.764**	1										
Exploration capabilities (3)	770	4.43	0.65	0.998	-1.223	.868**		1									
Technological innovation capability (TIC) (4)	770	4.11	0.52	1.163	-0.974	.715**	.343**	.600**	1								
Product innovation (5)	770	4.4	0.56	-0.072	-0.807	.647**	.293**	.715**	.676**	1							
Process innovation (6)	770	3.98	0.61	1.670	-0.950	.613**	.585**	.443**	.952**	.418**	1						
Competitive advantage (CA) (7)	770	4.16	0.51	0.201	-0.721	.642**	.562**	.502**	.663**	.629**	.629**	1					
Cost leadership strategy (8)	770	3.94	0.75	1.049	-1.393	.293**	.456**	.077*	.380**	.047	.448**	.808**	1				
Differentiation strategy (9)	770	4.38	0.61	0.994	-0.597	.720**	.388**	.749**	.650**	.702**	.510**	.694**	.135**	1			
Firm performance (FP) (10)	770	4.10	0.58	-0.207	-1.223	.578**	.289**	.620**	.503**	.646**	.352**	.478**	.127**	.648**	1		
Financial performance (11)	770	4.29	0.7	-0.427	-0.974	.503**	.189**	.587**	.402**	.646**	.227**	.372**	.020	.600**	.919**	1	
Growth performance (12)	770	3.88	0.6	0.563	-0.807	.523**	.349**	.494**	.505**	.470**	.428**	.495**	.241**	.538**	.837**	.553**	1

* $p < .05$. ** $p < .01$.

showed an ideal two-dimensional distribution upon examining the factor pattern for all items. (The detailed results of factor analysis with dimensions and the measures are given in the Supplemental File, Table A1).

Descriptive Statistics and Correlations

According to the results obtained from the study, the mean of the OA Scale was calculated as 4.38 ± 0.47 , the mean of the TIC Scale was 4.11 ± 0.52 , the mean of the CA Scale was 4.16 ± 0.51 , and the mean of the FP Scale was calculated as 4.1 ± 0.58 . The mean value of the scales was obtained by dividing the sum of the sample data by the sample volume and representing a single value that best represents the sample. The Standard Deviation value is obtained by taking the square root of the mean of the squares of the deviations from the mean. This value indicates the closeness of the data to the mean. According to the results obtained, our data show that it is close to the mean and has a homogeneous distribution. Moreover, when the kurtosis and skewness values of the scales and sub-dimensions were considered, they were found to be in the range of ± 2 . Skewness and kurtosis are two elements of normality (Tabachnick & Fidell, 2013, p. 79). According to George and Mallery (2010), when skewness and kurtosis values are in the range of ± 2 , the data are accepted to have a normal distribution (Eygü, 2018). Since the relationship between continuous variables has a normal distribution, it is shown in Table 2 with Pearson's correlation analysis. The means, SD, skewness and kurtosis are also presented in Table 2.

When the correlation table is examined, it was seen that the correlations of all overall scale scores and all sub-dimension variables were significant both among themselves and with each other. As shown in Table 2 with Pearson's correlation analysis, there was a statistically significant positive and high-level relationship between the OA Scale and the TIC Scale ($r = .715$; $p < .01$). Statistically significant positive and moderate relationship between the following scale was also found: the OA Scale and CA Scale ($r = .642$; $p < .01$), the OA Scale and the FP Scale ($r = .578$; $p < .01$), the TIC Scale and the CA Scale ($r = .663$; $p < .01$), the TIC Scale and the FP Scale ($r = .503$; $p < .01$), and the CA Scale and the FP Scale ($r = .478$; $p < .01$).

Analysis and Result

A series of simple and multiple linear regression analyses were conducted using SPSS 25 and SPSS PROCESS V.4 add-in to test our hypotheses. As shown in Table 3, we first regressed the direct hypothesis to provide more insight.

Table 3. Regression Analysis Results of Impact of Independent Variables on Dependent Variables.

H	Model	N	Standardized beta	F	Adjusted R ²	Result
H1	OA → FP	770	.578***	386.104	.334	Supported
H2	OA → CA	770	.642***	537.193	.411	Supported
H3	TIC → FP	770	.503***	537.193	.252	Supported
H4	TIC → CA	770	.663***	601.565	.439	Supported
H5	CA → FP	770	.478***	227.914	.228	Supported

Note. OA = organizational ambidexterity; TIC = technological innovation capability; CA = competitive advantage; FP = firm performance.

* $p < .05$. *** $p < .001$. VIF < 5.

Table 4. The Effect of the Mediation Variable According to Regression Analysis Results.

Hypothesis	H6 and H7		H8 and H9			
	Model 1 OA*TIC → FP	Model 2 OA*TIC*CA → FP	Model 3 OA*TIC → FP	Model 4 OA*TIC*CA → FP	Model 5 OA*TIC → FP	Model 6 OA*TIC*CA → FP
Sample Size(N)	770	770	386	386	384	384
Sector	All Sectors	All Sectors	Manufacturing Sector	Manufacturing Sector	Service Sector	Service Sector
Independent variable						
OA (β)	.447***	.400***	.485***	.467***	.327***	.201**
TIC (β)	.183***	.126**	.167***	.140**	.273***	.146*
Mediator						
CA (β)		.139***		.067		.307***
R ²	.351	.361	.359	.362	.321	.359
Adjusted R ²	.349	.358	.356	.357	.318	.354
F Value	207.402***	144.062***	107.232***	72.203***	90.169***	70.820***
Results	Partially supported		Supported			

Note. Independent Variables: OA = organizational ambidexterity; TIC = technological innovation capability; CA = competitive advantage; Dependent Variable: FP = firm performance.

* $p < .05$. ** $p < .01$. *** $p < .001$. VIF < 5.

Table 5. Hayes Test Results.

Hypothesis	N	Sector	Mediator variable	X, X1 and Y	Indirect effect	Boot SE	Boot LLCI	Boot ULCI	Mediation type	Result
H6 and H7	770	All Sectors	CA	OA, TIC and FP	0.0581	0.0231	0.0161	0.1065	Partial Mediation	Partially Supported
H8 and H9	386	Manufacturing Sector	CA	OA, TIC and FP	0.0251	0.0221	-0.0132	0.0724	No Mediation	Supported
	384	Service Sector	CA	OA, TIC and FP	0.1342	0.0408	0.0607	0.2189	Partial Mediation	

Note. Independent variables: organizational ambidexterity (OA) = (X); technological innovation capability (TIC) = (X1); competitive advantage (CA) = (M); dependent variable: firm performance (FP) = (Y).

H1 proposed that OA positively impacts FP ($\beta = .578$, $p < .001$, $R^2 = .334$). H2 proposed that OA positively impacts CA. ($\beta = .642$, $p < .001$, $R^2 = .411$). H3 predicted that TIC positively impacts FP. ($\beta = .503$, $p < .001$, $R^2 = .252$). H4 predicted that TIC positively impacts CA. ($\beta = .663$, $p < .001$, $R^2 = .439$). H5 proposed that CA positively impacts FP. ($\beta = .478$, $p < .001$, $R^2 = .228$). In simple linear regression tests, no multicollinearity problem was observed among the

independent variables ($VIF < 5$). The results indicated that OA and TIC are significant predictors when FP and CA are selected as dependent variables. Also, CA has a significant effect on FP. Therefore all five direct hypotheses, H1, H2, H3, H4, and H5 were supported.

Next, following the research design, multiple linear regression analysis was carried out to test the mediating role of CA in the effect of OA and technological innovation capabilities on FP. Baron and Kenny's model was

followed (Baron & Kenny, 1986). First, the effect of OA and TIC on FP was examined through Model 1. Then, to examine the effects of the mediation variable, CA was added to the regression model (Model 2) together with the independent variable and regression equations explaining a dependent variable were established using two independent variables. The relationship between OA and FP was significant for both Model 1 and Model 2. The standardized Beta coefficient of OA decreased from 0.447 to 0.400, and the standardized Beta coefficient of TIC was reduced from 0.183 to 0.126. In both multilinear regression tests, no multicollinearity problem was observed among the independent variables ($VIF < 5$). These results show that the CA partially mediates the relationship between OA and TIC on FP. Hence, hypotheses H6 and H7 were partially accepted.

Finally, we analyzed how the mediating role of CA in the effect of OA and TIC on FP varied by sector (Table 4). When the data were analyzed for the manufacturing sector (via Model 3 and Model 4) and service sectors separately (via Model 5 and Model 6). In the manufacturing sector, we first tested the Model 3. Then, we added the mediator variable into the model (Model 4), which the relationship between the dependent and independent variables was diminished. For both models (Model 3 and Model 4) relationships between OA and FP were significant. The standardized Beta coefficient of OA decreased from 0.485 to 0.467, and TIC was reduced from 0.167 to 0.140. However, the CA Beta coefficient (β) was not found to be statistically significant ($p > .05$) which shows us that for the manufacturing sector the relationship between OA and TIC on FP was not mediated by CA for the current study. In the service sector, we first tested the Model 5. Then, we added the mediator variable into the model (Model 6), which the relationship between the dependent and independent variables was diminished. The relationship between OA and FP was significant for both Model 5 and Model 6, and the standardized Beta coefficient of OA decreased from 0.327 to 0.201, and TIC was reduced from 0.273 to 0.146. These results show us that the CA partially mediates the relationship between both OA and TIC on FP in the service sector. In addition, in multiple linear regression tests, it was observed that there was no problem of multicollinearity between independent variables for all the test including Model 3, Model 4, Model 5 and Model 6 ($VIF < 5$). To sum up, the relationship between OA capability and TIC on FP was partially mediated by CA in the service sector, and it was not mediated by CA in the manufacturing sector. The results revealed a difference in the effect of the mediating role of CA. Since the difference was observed empirically, hypothesis H8 and H9 was accepted.

The Effect of the Mediation Variable was also examined by SPSS PROCESS V.4 add-in using Model 4. According to the test developed by Hayes, the indirect

effect of X on Y is an important indicator for the mediation variable. Results are given according to the confidence intervals. A significant mediation effect is mentioned if there is no zero value between BootLLCI and BootULCI values (Hayes, 2022). This model has no p-value (Sonmez Cakir & Adiguzel, 2020). In this test, level of confidence for all confidence intervals in the output: was 95.0000 and the number of bootstrap samples for percentile bootstrap confidence intervals was 5,000. As seen in Table 5, during the analysis, OA Independent variable(X), TIC Covariate(X), FP independent variable (Y) were taken as CA mediator (M) as required by Model 4. BootLLCI value (0.0161) and BootULCI value (0.1065) were found in the model where Hypothesis H6 and H7 was tested. There is no zero between the two values. CA has a partial mediation effect of OA and TIC on FP. For the analysis of the H8 and H9 hypotheses, the mediation effect was examined on a sectoral basis using the same model. BootLLCI value (−0.0132) and BootULCI value (0.0724) were found for the manufacturing sector. There is a zero between the two values. Therefore, there is no mediation effect. BootLLCI value (0.0607) and BootULCI value (0.2189) were found for the service sector. The values between these two values are not zero again. CA has a partial mediation effect of OA and TIC on FP. All results supported the results of multiple regression analysis to determine the mediation effect based on the Baron and Kenny approach.

Discussion and Implications

The main aim of this study was to reveal the mediating role of CA in the effect of OA and TIC on FP in a developing country. The results showed that the positive and statistically significant effect of OA and technological innovation capabilities on FP is partially mediated by CA, and this mediation effect differs according to sectors.

It is observed that OA affects FP at a rate of 33.4% and CA at a rate of 41.1% alone. Similarly, TIC affects FP at a rate of 25.2% and CA at a rate of 43.9%. The positive effect of OA on FP (hypothesis H1) supports the studies by Gibson and Birkinshaw (2004), He and Wong (2004), and Han and Celly (2008). The results are also consistent with the conceptual study by Raisch and Birkinshaw (2008) and the meta-analysis results obtained by Simsek (2009). However, they differ from the meta-analysis findings in which Junni et al. (2013) revealed that OA was particularly important for performance in non-manufacturing industries and at higher levels of analysis. Since the current study found that the effect of OA on FP was not more substantial in the manufacturing industry, the findings are compatible with Junni et al. Also, this study's results contrast the Altındağ and Bilaloğlu

Aktürk (2020) study. The positive effect of OA on CA (hypothesis H2) is consistent with the studies by Hamel and Prahalad (1993), Tushman and O'Reilly (1996), Clauss et al. (2021), Han and Celly (2008), Amniattalab and Ansari (2016), Jurksiene and Pundziene (2016), and Bakan and Sezer (2016). The positive effect of TIC on FP (hypothesis H3) supports the studies by He and Wong (2004), J. Guan and Ma (2003), Yam et al. (2011), Yang (2012), Azubuike (2014), Chen et al. (2020) and, YuSheng and Ibrahim (2020). The results are also consistent with Camisón and Villar-López (2014), in which technological capabilities for products and processes lead to superior FP and Ferreira et al. (2020), which found a positive relationship between innovation capability and FP and CA. The positive effect of TIC on CA (hypothesis H4) supports the empirical studies by Yam et al. (2004), J. C. Guan et al. (2006), Liu and Jiang (2016), and Strand et al. (2017). The results are consistent with the statement of Freeman (1994) that the primary source of CA is TIC and support the Betz (2003) and Saunila (2020) studies. The positive effect of CA on FP (hypothesis H5) was founded at 22.8% and revealed similar results to the studies by Albaumb and Tse (2001), Akman et al. (2008), and Peker et al. (2016). And opposite results to Pelham and Wilson (1995) and Chan et al. (2004), and Ma (2000).

OA and TIC are relatively new concepts that have been gaining popularity recently. When these two independent variables are included in the model together, it was seen that the rate of disclosure of FP increases to 34.9%. This effect increases even more to 35.8% in the presence of CA. This result shows that these three variables work harmoniously for the selected sectors, and the model is well constructed. Since the number of studies in the literature on these variables is currently limited, our findings that CA plays a partially mediating role in the effect of OA (hypothesis H6) and TIC (hypothesis H7) on FP are important. This finding indicates that in the conceptual framework presented, CA has a partially mediating effect, as well as other variables mediating this relationship. Thus, while making a theoretical contribution to the literature, this relationship also shows a direction for the existence of other mediating variables for future research.

Our analyses also revealed that the mediating role of CA in the effect of OA and TIC on FP differed in the manufacturing and service sectors (hypotheses H8 and H9, respectively). CA was found to significantly mediate both capabilities on FP in the service sector but not in the manufacturing sector. On the other hand, when the model is analyzed by sector, it shows that the disclosure rate of FP increased from 35.6% to 35.7% for the manufacturing sector and from 31.8% to 35.4% for the service sector. Although CA has not mediated in the

manufacturing sector, these numerical expressions show the importance of these capabilities on FP.

This intersectoral difference can be explained by the high level of current technological innovation and institutionalization in the information technology sector, which was selected to represent the service sector in this study. As known from the literature studies, the textile industry is a low-technology-intensive industry, and the level of institutionalization is low. The inability of the institutional structures of the textile/apparel/leather firms in the manufacturing sector to respond to organizational skill requirements as quickly as the information technology firms in the service sector can be explained by the nature of the business in this sector.

Limitations and Future Research

The present study has some considerable limitations. The findings are based on the results of a single study, and therefore, iterative studies can be conducted to perform more detailed analyses on firms with varying technology intensities. From an empirical perspective, our approach used a limited sample size. Since the research was conducted in two sectors, textile/ready-made garment/leather firms representing the manufacturing sector and information technology firms representing the service sector, the research findings are limited to these two sectors. Our study can be repeated with datasets from different samples in larger research populations to generalize the results. The current study was conducted in Turkey, considered a developing country. It is recommended that the study be replicated in other countries at the same level of development to increase the generalizability of the results. In addition, differences can be revealed by working in countries with different development levels.

Furthermore, a comparison can be made on whether the manufacturing and service sectors studied differ. Researchers who want to work on this subject are encouraged to examine firm behaviors in different samples by increasing the number of sectors. Moreover, concerning future research, our findings indicate that more industry-based studies are needed to show the complexity and heterogeneity of causal chains of relationships leading to the same outcome. This is especially valid for performance studies. In addition, the fact that the conceptual framework as a mediating variable is not only modeled with a CA is another limitation that provides a significant opportunity for researchers. Future researchers can test the same model with different mediating variables. Another major research perspective would be to go deeper to understand why such configurations of national incentive system design, the role of standards, and overall performance can occur concerning

organizational agility, organizational absorptive capacity, and innovation capability. To this end, conducting empirical iterations and qualitative studies is recommended. This may be appropriate for research into the nature of cause-effect relationships and may provide a better understanding of the relationships between these antecedent conditions and performance outcomes. In addition, since our study is limited to data collection tools, it is recommended to conduct mixed studies that combine quantitative and qualitative research on this issue.

Conclusion

The strategic management literature reveals the vital importance of the concepts of OA and TIC for FP. However, there are few versatile studies on these concepts. In particular, we have not found a study like our research on the mediating role in the literature. First, the current study separately presented the effects of OA and TIC on FP and CA. Then, the effect of CA on FP was demonstrated. The mediating role of CA in the effect of OA and TIC on FP was presented empirically to answer the fundamental research questions. Afterward, the research conducted simultaneously and in comparison with the sector showed empirical findings that the mediating role of CA in the effect on FP differed in the manufacturing and service sectors. The most important theoretical contribution of the study is to show businesses in different fields in the region in numerical terms the type of organization and innovation they need to increase their capabilities in the context of the RBV. This finding shows the importance of OA and TIC for businesses. It also provides empirical evidence of this ability to increase CAs and FP. Finally, in managing FP, we drew attention to the fact that conducting research on OA and TICs, in addition to CA, was a well-modeled research flow.

Managers of the business world are faced with the conditions of fierce competition. It is crucial for firms that will be the future winners to include these two essential capability sets in their strategic plans to make them their core competencies. OA capability, which allows firms to establish balance and switch between their capabilities, is effective in obtaining CA and providing sustainability. In this way, the FP, which shows a positive increase, will strengthen the firms both financially and non-financially and will extend their survival times. Furthermore, in today's world, where change has accelerated significantly, OA capability will contribute to the adaptation to the external environment by providing firms with the capability to respond quickly to the situations they encounter.

The findings of our study offer contributions to all business managers and policymakers in developing countries. The positive mediation effect of the CA of the business also provides positive contributions to the development of the country in which it is located due to the contribution it provides to the performance of the business. In order to increase FP in developing countries, essential policy recommendations should be designed to facilitate the achievement of OA and the acquisition of TIC for businesses. Especially TIC is regarded as one of the cornerstones of economic development, on which not only private firms but also governments, states and their associations and non-governmental organizations work nowadays. Firm managers who want to succeed always have to display high performance as active players in the competition. From a managerial perspective, it is crucial for managers of these firms, which naturally need to have a CA, to be aware of the organizational requirements that make technological innovation possible. Establishing organizational structures that will pave the way for innovation in the future race is essential. Innovative efforts made with this capability set enabling the market offering of value propositions to meet the needs of today's consumers through products/services and processes will play an active role in creating and maintaining CA and thus making the FP superior by always obtaining high profits, thus creating returns that will extend the life of firms.

Acknowledgments

The authors thank editors and anonymous reviewers for their kindest help and guidance.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

An Ethic Statement

This article was produced from the doctoral thesis prepared by the 1st author under the supervision of the 2nd author. For her doctoral dissertation titled "The Mediator Role of Competitive Advantage in the Effect of Organizational Ambidexterity and Technological Innovation Capability on Firm Performance," Istanbul Beykent University Permission was obtained from the

Scientific Research and Publication Ethics Committee for Social and Human Sciences, dated 20.12.2021 and numbered 134.

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Supplemental Material

Supplemental material for this article is available online.

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